
UNIT 29 BEHAVIOURAL FINANCE

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29.0 OBJECTIVES

After studying this unit, you will be able to:

- explain the concept of behavioural finance;
- describe the departure from traditional finance concepts to behavioural finance;
- differentiate between traditional finance and behavioural finance;
- explain the Efficient Market Hypothesis;
- give a behavioural explanation to important market puzzles;
- explain the causes of irrational investor behaviour; and
- describe the heuristics and biases that explain it.

29.1 INTRODUCTION

Traditionally, finance has been based around the concept of “efficient markets” which states that the price of an asset or security is ‘right’, i.e., the price reflects its fundamental value and all the available information. For a long time, financial theory was governed by the idea that investors act rationally as they have all the available information and they aim to maximize profits. The theory believes that even if there is any ‘noise’ by irrational investors in the market, it is balanced by rational investors. Traditional finance has been surrounded by three unbounded traits (rationality, self-control and self-interest) which form a base for ‘homo economicus’.

However, over time, it was felt that the traditional theory was not able to explain various market anomalies like market volatility and unpredictability in the market, market crashes, irrational investor behaviour, stock market bubbles etc. Traditional finance also was not able to address questions like ‘why does an investor trade, how does an investor trade, what are the factors that help an investor decide his portfolio etc.

These market anomalies and investor behaviour were explained by an alternative approach termed ‘Behavioural Finance’ which focuses on cognitive, sociological and psychological influences on the decision-making of investors and financial markets. It focuses on other important aspects of the decision-making process, i.e., how people invest, what they value and how they adjust for risk by embracing findings from psychology and sociology. Behavioural finance highlights that the investors are not always rational and the prices are not always efficient.

Over the last few decades, behavioural finance has become an extremely popular field of study as an application in business and finance. It centres around the fact that investors’ decisions are influenced by their own biases. Thus, investors behave quite differently from ‘homo economicus’ and their financial choices are guided by their social and psychological needs. It also highlights that investors as humans have limits to self-control and often apply heuristics to make complex decisions.

We have discussed traditional finance and the ‘Efficient Market Hypothesis’ (EMH) in previous units. In the present unit, we will discuss what is beyond traditional finance and EMH. In this unit, we will discuss behavioural finance, the factors that led to the growth of this parallel field and how it attempts to explain investors’ behaviour and financial markets.

29.2 CONCEPT OF BEHAVIOURAL FINANCE

As discussed in the previous section, behavioural finance focuses on understanding the thought process of investors and how it impacts their decisions. It attempts to explain the degree of various influences on a

person's decision-making. It further explains various market anomalies and heuristics and biases that cause it.

Behavioural finance became popular after 2002 when Daniel Kahneman was awarded the Nobel Prize for his work on psychological research into economic science, especially concerning human judgment and decision-making under uncertainty. Behavioural finance is studied in two aspects: individual investors, and the whole market. Accordingly, behavioural finance is classified into two aspects: (i) microbehavioural finance, and (ii) macrobehavioural finance.

Microbehavioural finance deals with the cognitive, psychological and sociological influences on the individual investors. It delves into behavioural biases that distinguish them from rational economic beings. It also studies heuristics as rules of thumb that lead to suboptimal decision-making.

Macro-behavioural finance, on the other hand, examines market anomalies that make the market inefficient. It focuses on the reasons behind stock bubbles, market volatility and limited arbitrage. This unit deals with both aspects of behavioural finance.

Check Your Progress 1

1) What is meant by behavioural finance?

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2) What are the assumptions of traditional finance theory?

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3) Which one of the following is not a trait of 'homo-economicus'?

- a) Rationality
- b) Self-control
- c) Cognitive bias
- d) Self-interest

4) Microbehavioural finance deals with:

- a) Irrational investor Behaviour
- b) Stock market volatility
- c) Bubbles
- d) Limited arbitrage

29.3 DIFFERENCE BETWEEN TRADITIONAL FINANCE AND BEHAVIOURAL FINANCE

Traditional finance centres around the concept of investor and market rationality. It suggests that given the available information, investors make financial decisions and in the course of decision-making, are not affected by emotions. Traditional finance does not consider the role of heuristics and accordingly does not contemplate investors being affected by any bias. On the other hand, behavioural finance examines the role of psychology and sociology in financial decision-making. It considers the influence that mood, fear, overconfidence, past experiences etc. have on investors when they decide to devote funds to any asset.

More than contrasting, behavioural finance provides an alternative approach to traditional finance to have a better understanding of the market and investors. Behavioural finance draws from real life that investors are sometimes irrational, and biased and are influenced by various social and psychological factors. For example, it is often observed that while buying stocks, many times investors follow other investors because they think that everyone else knows more and makes better decisions. This reflects a herding mentality and can lead to sub-optimal decisions by an investor.

The main points of contrast between traditional and behavioural finance are:

- Traditional finance centres around the idea of investor rationality and an unbiased approach to processing information for decision-making.

While behavioural finance considers the influence of various biases in investment decision-making, it recognizes that human emotions and other psychological influences may make an investor irrational. Overconfidence, familiarity bias, herding mentality, etc. deter the decision-making capacity of an investor.

- Traditional finance assumes the availability of complete information to everyone in the market. It further assumes that investors examine all the information rationally before making a decision. Thus, investors always make the right decision. Behavioural finance, on the other hand, centres around the idea of bounded rationality, i.e., investors do not process all the information and they try to satisfice instead of maximise. Even if the information available is accurate, investors will still make errors in judgment.
- Traditional finance centres around the 'efficient markets hypothesis', i.e., markets are efficient and the price of a stock always reflects its fundamental value. It is based on the availability of complete knowledge and investor rationality. However, behavioural finance disagrees with the fact that markets are efficient and discusses various anomalies that lead to market volatility and cannot be explained through EMH.

- Investors, according to traditional finance have perfect self-control and they do not let any other aspect of their past or personality influence their decision-making. Behavioural finance considers self-control problems in investors and examines how it affects decision making in turn affecting the whole market.
- Classical finance theory asserts that any noise is balanced out by rational traders but according to behavioural finance, there is a limit to arbitrage and prices are not always right and may stay in a non-equilibrium state for certain periods.

29.4 GROWTH AND ORIGIN OF BEHAVIOURAL FINANCE

The cognitive and psychological underpinnings to explain stock market Behaviour can be traced back to the 60s and 70s also but it was mainly in the 1980s when Behavioural finance evolved with an attempt to explain stock market bubbles and other market anomalies. Since then, foundational ideas of Behavioural Economics have been used to explain the reason behind these anomalies. The ideas of Daniel Kahneman and Amos Tversky have played a major role in explaining the contradictions in the traditional theory of finance.

Over time, behavioural finance has developed through two phases: (i) through examples of market anomalies and deviation from standard finance assumptions; and (ii) through explanation of these anomalies. The decade of the 1980s saw various important concepts like ‘disposition effect’ and ‘equity premium puzzle’ being developed by various economists. In 1981, Robert James Shiller discussed market anomalies and pointed out how rational factors alone cannot explain stock market volatility. De Bondt and Richard Thaler’s paper on stock market overreaction in 1985 elaborated on value anomaly. We will discuss all these concepts later in the unit.

The late 1990s and the early 2000s saw behavioural finance becoming part of mainstream finance. It was able to explain the reasons behind many market anomalies. Many concepts were developed during this time such as (i) limit to arbitrage, (ii) myopic loss aversion, (iii) over-reactions, and under-reactions of the market. This decade also saw research in the area of irrational investors and the behavioural mistakes they make. Robert Shiller's famous book ‘Irrational Exuberance’ in 2000 rightly predicted the crash of the dot-com stock market bubble and spread the ideas of behavioural finance as a parallel field of traditional finance, an understanding of which could increase the predictability of the stock market. Around the same time, Andrei Shleifer also worked to explain how noise traders in the financial market lead to limited arbitrage and cause market anomalies. Further, the Nobel Prize for two consecutive years (2001 and 2002) increased the popularity and acceptance of behavioural finance. You may be aware that the Nobel Prize was awarded in 2001 to George Akerlof, Michael Spence, and Joseph Stiglitz

“for their analyses of markets with asymmetric information” and to Daniel Kahneman in 2002 for “having integrated insights from psychological research into economic science, especially concerning human judgment and decision-making under uncertainty”.

As of now, the field of behavioural finance is still evolving and is a popular research area. The psychological, cognitive and sociological foundations of financial decision-making are still being explored to have a better understanding of the markets. However, behavioural finance is criticized for being “Micro-efficient” and not “Macro-inefficient”, i.e., it can explain individual behaviour of investors effectively but is not very useful in predicting the overall market. We can explain market anomalies through behavioural finance in the long run, but if we try to predict variations in the stock market in the short run we will not be very successful. The field is still growing, with more and more research to understand sources of irrational behaviour and market anomalies.

Check Your Progress 2

- 1) Compare and contrast traditional finance with behavioural finance.

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- 2) Discuss the growth and evolution of behavioural finance.

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- 3) Traditional theory assumes that investors are _____
whereas behavioural finance considers investors to be _____

- a) Rational, rational
- b) rational, irrational
- c) irrational, irrational
- d) irrational, rational

- 4) Which one of the following is not true about behavioural finance?
- a) It assumes the availability of complete information to everyone in the market.
 - b) It considers issues such as self-control in investors and examines how it affects decision making which in turn affects the whole market.
 - c) it centres around the idea of bounded rationality.
 - d) There is a limit to arbitrage, and prices are not always right and may stay in a non-equilibrium state for certain periods.

29.5 EFFICIENT MARKETS HYPOTHESIS AND ANOMALIES

29.5.1 Efficient Market Hypothesis and its Critique

We have already discussed the concept of ‘efficient markets hypotheses’ (EMH) in Unit 25. According to EMH, market prices fully reflect all available information. The EMH is one of the important cornerstones of traditional finance theories. It was developed independently by Paul A. Samuelson and Eugene F. Fama in the 1960s and has been applied extensively in traditional finance. In an efficient market, all relevant information is freely available to profit-maximizing rational investors who are trying to predict the value of various securities. According to EMH, markets are rational and prices of stocks fully reflect all available information. Since such information is readily available, the price of securities quickly adjusts to the change. The idea behind EMH is that when some new information arises or any event occurs, the news spreads very quickly and is incorporated into stock prices. For example, if a currency note is lying on the road, it will not be there for long as it will be surely picked up by someone. Similarly, all information is picked up by market agents and used in decision-making. Current prices are considered to be the best approximation of a company’s intrinsic value.

The EMH is associated with the idea of a “random walk”. According to the random walk theory, past movements or trends of a stock price (or financial market) cannot be used to predict its future movement and it is not possible to outperform the market without assuming additional risk. The logic behind the random walk theory is that stock prices reflect information or news about the stocks. Since the news is unpredictable, today’s situation or price cannot be used to predict future values. As such, experienced investors do not have any added advantage over uninformed investors.

Behavioural finance, on the other hand, believed in ‘momentum’ in short-run stock prices. According to behaviour finance, investors follow market trends, i.e., if they notice a rise in the price of a stock, they invest in the market in a

kind of “bandwagon effect.” leading to sometimes what is called ‘irrational exuberance’, i.e., investor enthusiasm that leads to rising in a stock price higher than its fundamental value. Another rationale for ‘momentum’ as given by behavioural finance is the investor’s under-reaction to the new information. Investors take time to change their existing perception or understanding of a particular stock. Thus, the full impact of any news or announcement is realized over a period leading to a positive correlation in prices (thus, changes in prices are not random)

Another significant critique of behavioural finance was regarding assumptions about human Behaviour. They critiqued the rationality assumption by classical finance and focused on sociological, psychological and cognitive influences leading to behavioural anomalies. Behavioural finance documented various departures from a classical paradigm in the form of behavioural biases that guide investor behaviour especially while deciding under uncertainty. Many times, these biases lead to irrational behaviour and ultimately to sub-optimal outcomes in the market. Some of the biases that we will discuss in this unit are herding, overconfidence, overreaction, loss aversion, and mental accounting which dominate the thinking process of an investor. These biases in turn lead to market anomalies such as the ‘equity premium puzzle’ (Stocks have earned disproportionately higher returns over the years yet investors are relatively unwilling to hold them), the ‘volatility puzzle’ (Stock returns are highly variable with large dispersions) and the ‘predictability puzzle’ (Stock returns are predictable suggesting that persistent profits are not being eroded by arbitrage).

29.5.2 Market Anomalies

Anomalies are occurrences that deviate from the predictions of the efficient markets hypothesis and undermine its core assumptions. (Tversky & Kahneman 1986) defined market anomalies as “an anomaly is a deviation from the presently accepted paradigms that are too widespread to be ignored, too systematic to be dismissed as random error and too fundamental to be accommodated by relaxing the normative system”.

Anomalies can be of three types: (i) Fundamental Anomalies, (ii) Technical Anomalies, and (iii) Calendar or Seasonal Anomalies

i) Fundamental Anomalies

- Value Anomaly: It refers to the over-estimation of future earnings and returns of growth companies.
- Low Price to Book: It is often noticed in the stock market that the stocks having a high book-to-market ratio generate less return than the stocks with a low price-to-book ratio.
- High Dividend Yield: There is a positive correlation between yield and stock return. Stocks with high dividend yields outperform the market and generate more returns.

- **Neglected Stocks:** The prior neglected stocks generate more return subsequently over some time. Accordingly, the prior best performers underperform the index.
- **Low Price to Earnings (P/E):** The stocks with low price-to-earnings ratios are likely to generate more returns and outperform the market, while the stocks with high price-to-earnings ratios tend to underperform in the market.

ii) Calendar Anomalies

Calendar anomalies reflect an irregular pattern of stock returns in various periods, i.e., movement in stock prices from day to day, month to month, year to year etc. Some of the Calendar anomalies are as follows:

- **Weekend Effect:** The stock prices are likely to fall on Monday, i.e., the closing price of Monday is less than the closing price of the previous Friday.
- **Turn-of-the-Month Effect:** It reflects the tendency of stock prices to increase in the last trading day of the following month, and the first three days of next month.
- **Turn-of-the-Year Effect:** This anomaly describes the increase in the prices of stocks and trading volume of the stock exchange in the last week of December and the first half month of January.
- **January Effect:** The tendency of small-company stocks to generate more return than another asset in the first two to three weeks of January is called the January effect.

iii) Technical Anomalies

Various techniques can be used to forecast future stock prices based on past prices and relevant information in the market. Traditionally, researchers asserted that the market holds weak form efficiency, prices already reflect all the information and everyone has all the information. Thus, an investor cannot beat the market by earning abnormal returns. But various market anomalies deviate from the traditional viewpoint:

- **Moving Averages:** Buying and selling signals of stocks can be generated by long-period averages and short-period averages. A purchase signal is created when short-period averages rise over long-period averages. The best time to sell is when short-period averages fall below the long-period averages.
- **Trading Range Break:** A breakout is a potential trading opportunity that investors apply to make huge profits. This technique involves identifying support and resistance levels along with current price trends to identify possible entry and exit points. By buying or selling at the right time, investors can make huge profits.

29.5.3 Market Puzzles and Behavioural Explanation

Over time, Behavioural Finance has attempted to explain various market anomalies and puzzles. This section discusses about few popular puzzles in the stock market and their Behavioural explanation.

i) Equity premium puzzle

The equity premium puzzle, first documented by Mehra and Prescott, refers to the empirical fact that stocks have greatly outperformed bonds over the last century. Stocks have earned disproportionately higher returns than bonds, however, people still buy bonds. The level of risk aversion required for such a choice by investors has to be very high. Thus, the equity premium puzzle cannot be explained in terms of risk aversion. Behavioural finance explains this puzzle in terms of 'Prospect Theory'. The first explanation is provided in terms of 'loss aversion', i.e., investors are more sensitive to losses than to gains. Secondly, investors are myopic, i.e., investors are assumed to evaluate their portfolio frequently even if they are saving for their retirement. Thus, myopic loss aversion explains the equity premium puzzle. Another explanation for the puzzle is provided through 'ambiguity aversion'. Ambiguity aversion is a preference for known risks over unknown risks. Since the probability distribution of equity returns is unknown, people overweigh the probability of returns from bonds.

ii) Volatility Puzzle

The volatility puzzle reflects the high variability of stock prices having large dispersions. It is observed in the stock market that changes in stock prices are disproportionately higher than the changes in fundamentals. This observation is against the theory of EMH according to which, prices always represent the fundamentals of the assets.

Behavioural finance attempts to solve this puzzle through various biases that investors have. Various cognitive, social and emotional biases lead to sub-optimal decision-making by the investors which in turn affect the stock market and stock prices. It is noticed that while making a decision, investors overweigh more recent information, and consensus beliefs and seek confirming evidence which makes them under-react or overreact in the market.

Investors also tend to focus more on similar pieces of information called focal points which also affect their decision making. The reason behind the steep fall in prices rather than the upward correction is that people are loss-averse and tend to overweigh negative information and feedback.

iii) Predictability Puzzle

This puzzle states that stock returns are predictable from price-dividend ratios. Behavioural finance is explained in terms of behavioural biases like the disposition effect (the tendency to prematurely sell assets that have made